

Programming Fundamentals I

An Introduction to Structured Programming

Tarrant County College Department of Computer Science

COSC 1436

Version 0.1 (Draft) · 2025

This work is licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

Contents

Preface	vii
I Foundations	1
1 Computers, Data, and Programs	3
2 The Software Development Process	5
II Core Programming Concepts	7
3 Data Types, Variables, and Constants	9
4 Input, Output, and Operators	11
5 Selection	13
6 Iteration	15
III Organizing Code and Data	17
7 Functions and Procedures	19
8 Arrays	21
9 Strings	23
IV Writing Good Software	25
10 Designing for the User	27
11 Testing and Debugging	29
12 Version Control in Practice	31

Cambridge Pseudocode Quick Reference	33
---	-----------

List of Figures

Preface

Part I

Foundations

Chapter 1

Computers, Data, and Programs

Chapter 2

The Software Development Process

Part II

Core Programming Concepts

Chapter 3

Data Types, Variables, and Constants

Chapter 4

Input, Output, and Operators

Chapter 5

Selection

Chapter 6

Iteration

Part III

Organizing Code and Data

Chapter 7

Functions and Procedures

Chapter 8

Arrays

Chapter 9

Strings

Part IV

Writing Good Software

Chapter 10

Designing for the User

Chapter 11

Testing and Debugging

Chapter 12

Version Control in Practice

Cambridge Pseudocode Quick Reference
